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ORIGINAL RESEARCH ARTICLE

Microstructure and Properties of 316L Stainless Steel Fabricated by Ultrasonic Impact and External Wire Feeding-Assisted Tandem Wire and Arc Additive Manufacturing

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While tandem wire and arc additive manufacturing offers a higher deposition efficiency, excessive heat input can compromise forming quality and performance. To tackle this challenge, this paper introduces an external wire feeding-assisted tandem wire–arc additive manufacturing system that minimizes thermal accumulation during the manufacturing process. Furthermore, it incorporates ultrasonic impact to refine grain structure and enhance mechanical properties. Our findings indicate that, for stainless steel 316L, the efficiency of this external wire feeding-assisted system reaches 9.75 kg/h. The application of ultrasound effectively inhibits the formation of coarse columnar dendrites, modifies grain growth patterns, and achieves grain refinement. Additionally, the ultrasonic impact not only increases the number of grain boundaries but also elevates the dislocation density within the material, thereby hindering dislocation movement and enhancing the overall hardness and tensile strength of the fabricated components. After the introduction of ultrasonic impact, the average hardness of the additively manufactured samples increased from 183.9 to 190.8 HV, while the transverse tensile strength rose from 495 ± 7.8 to 526 ± 9.4 MP.

Keywords external wire feeding, mechanical properties, microstructure, tandem wire–arc additive manufacturing, ultrasonic impact

1. Introduction

Metal additive manufacturing (MAM) is primarily categorized into two main types based on the forming method: directed energy deposition (DED) (Ref 1) and powder bed fusion (PBF) (Ref 2-6). Directed energy deposition (DED) is a method for melting materials layer by layer, where metal wire or powder is delivered simultaneously with the energy needed to melt it, thereby forming small melt pools and continuously depositing material layer by layer. Wire–arc additive manufacturing (WAAM) is a type of directed energy deposition (DED). In recent years, wire–arc additive manufacturing (WAAM) technology (Ref 7-9) has rapidly developed due to its advantages of low cost, high efficiency, strong flexibility, and short production cycles (Ref 10, 11). Based on the type of arc, WAAM can be classified into gas tungsten arc additive manufacturing (GTA-AM) (Ref 12), gas metal arc additive manufacturing (GMA-AM) (Ref 13), and plasma arc additive manufacturing (PA-AM) (Ref 14). Among these, GMA-AM holds broad prospects in the direct manufacturing of large-scale metal structures owing to its high deposition efficiency. However, with the increasing demand for large and super-large structures, there is a need to further enhance the efficiency of GMA-AM. Consequently, multi-wire multi-arc additive manufacturing processes have received extensive attention (Ref 15, 16), and related research, represented by tandem wire and arc additive manufacturing, has been carried out.

For example, Queguineur et al. (Ref 17) introduced a dual-wire cold metal transfer (CMT) arc additive manufacturing process for nickel aluminum bronze (NAB) components, which significantly increased the deposition efficiency from 2-3 to 7-10 kg/h compared to single-wire CMT arc additive manufacturing. This demonstrates the potential for improving productivity in the production of large components. Martina et al. (Ref 18) used a dual-wire additive manufacturing method to produce martensitic stainless steel 17-4 pH, achieving a deposition efficiency of 9.5 kg/h—double that of traditional methods—with no significant defects and hardness values comparable to the base material. Xue et al. (Ref 19) optimized the forming process of 30-layer stainless steel components deposited using dual-wire dual-arc pulsed CMT (CMT+P) technology. Although this process resulted in increased heat accumulation, leading to grain coarsening and a slight decrease in mechanical properties, it achieved a maximum deposition rate of 5.4 kg/h, with both strength and elongation meeting the basic standards for stainless steel forgings. Wu et al. (Ref 20) proposed a dual-arc additive manufacturing process based on Gas Metal Arc Welding (GMAW) for the production of 316L stainless steel (SS316L). In this process, the deposition rate of the dual-arc process was 1.25 times that of the single-arc

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process at the same power level, reaching 1.15 kg/h, while hardness and strength were also improved. It can be seen that while dual-wire dual-arc additive manufacturing increases efficiency, it often comes at the cost of increased heat input, which may lead to defects such as grain coarsening and stress deformation in the formed components (Ref 21). This can result in material anisotropy and reduced mechanical properties of the components, thereby limiting the application and promotion of dual-wire dual-arc additive manufacturing technology.

To mitigate heat input, common methods include employing pulsed arcs (Ref 22), external cooling (Ref 23), introducing low-temperature gases (Ref 24), and feeding additional wire (Ref 25). Among these, feeding additional wire is more convenient and can improve deposition efficiency while reducing heat input. For example, Xiong et al. (Ref 26) proposed a dual-wire GTA additive manufacturing technique, where extra wire is fed into the stainless steel GTA additive manufacturing melt pool. The additional wire consumes excess energy in the overheated melt pool, thereby enhancing wire melting efficiency and reducing thermal damage and heat accumulation in the deposited layers. Compared to single-wire GTA additive manufacturing, this method can increase forming efficiency by 0.5 times and enhance the tensile strength of stainless steel components by 16.14%. Han et al. (Ref 27) introduced a tandem wire GMA additive manufacturing technique, where a second wire is fed into the GMA process to consume arc energy, thus improving wire melting efficiency. Research indicates that the forming efficiency of tandem wire GMA additive manufacturing is 1.65 times that of traditional arc additive manufacturing, and thermal damage in the deposited layers can be reduced by 28.6%. Wang et al. (Ref 28) proposed a novel wire–arc directed energy deposition (DED) process that combines gas metal arc (GMA) welding with external cold wire feeding to achieve a high deposition rate and low material re-melting, with a maximum deposition rate reaching 14 kg/h. The addition of the cold wire significantly reduces re-melting and improves mechanical properties to some extent. Therefore, the introduction of the external wire can enhance efficiency while reducing the heat of the dual-wire dual-arc process, showing great application prospects.

Meanwhile, to further enhance the performance of additively manufactured components, ultrasonic impact treatment (UIT) can be introduced (Ref 29). By applying UIT to the molten pool, the metal surface is continuously impacted to refine the grain structure and improve its mechanical properties (Ref 30). Additionally, this technology can redistribute residual stresses in the deposited layers, thereby enhancing the quality of arc additive manufactured components. Su et al. (Ref 31) investigated the effects of different UIT methods on the microstructure and mechanical properties of 18Ni-300 steel arc welded components. The results showed that applying UIT could refine grain size, alter grain growth direction, and reduce anisotropy, leading to significantly improved material properties compared to those without ultrasonic treatment. Diao et al. (Ref 32) demonstrated that supplementing the arc additive manufacturing process of stainless steel with UIT could transform the austenitic stainless steel surface into martensite due to deformation, thereby improving the microstructure and enhancing the overall mechanical properties of the formed components. Chen et al. (Ref 33) found that applying ultrasonic assistance in metal inert gas (MIG) welding of aluminum alloys was more stable than traditional MIG welding, with significant improvements in weld penetration depth and droplet transfer in ultrasonic vibration-assisted MIG welding. Wang et al. (Ref 34) studied the effects of UIT on the microstructure and mechanical properties of wire–arc additive manufacturing (WAAM) 2219 aluminum alloy. The results indicated that UIT could refine grain size, reduce porosity, increase dislocation density, improve surface residual stress, and ensure uniform distribution of precipitates in the matrix, thereby effectively enhancing the mechanical properties of WAAM components.

Drawing upon the aforementioned research context, this paper introduces an innovative tandem wire–arc additive manufacturing technique that integrates ultrasonic impact treatment with external wire feeding. With 316L stainless steel as the primary material under investigation, the study is geared toward mitigating thermal accumulation, enhancing the productivity of arc additive manufacturing processes, and upgrading the properties of the resultant components. A comprehensive analysis is conducted to compare the macro morphology, microstructure, and mechanical properties of parts fabricated both with and without the application of ultrasonic impact. Furthermore, the deposition efficiency is assessed and benchmarked against other additive manufacturing methodologies, offering valuable insights into enhancing the efficiency and performance of arc additive manufacturing while also optimizing the mechanical attributes of the produced parts.

2. Experimental Methodology

Figure 1 illustrates the schematic of the ultrasonic impact-assisted and external wire feeding tandem arc additive manufacturing system. This system primarily consists of a tandem wire welding power source (SAF brand from France) and its accompanying two wire feeding mechanisms to provide dual-arc energy and wire feeding, an independent external wire feeding mechanism, an ultrasonic impact device and its power supply, a working platform, and a gas delivery system. In the actual additive manufacturing process, the tandem wire welding power source generates two arcs between the dual wires and the substrate or molten metal to melt the wire and form a molten pool. The external wire feeding mechanism synchronously feeds the wire into the molten pool located between the two arcs, utilizing the energy of the molten pool to melt the wire. The ultrasonic impact gun is positioned 5 cm behind the additive manufacturing gun and moves forward synchronously, acting on the surface of the molten metal at high temperature. The distance between the conductive nozzles of the two additive manufacturing guns is 20 millimeters, and their positions are adjustable. The angle between the additive manufacturing gun and the vertical direction is set at 15°, and the angle between the external wire and the plane of the two additive manufacturing guns is also set at 15°, the specific arrangement of the additive manufacturing guns and ultrasonic impact gun is shown in Fig. 2.

The experiment employed 316L stainless steel wire with a diameter of 1.2 mm, and the substrate for the additive manufacturing was 304 stainless steel, measuring 300 mm × 200 mm × 20 mm. Their chemical compositions are presented in Table 1. The process parameters for the experiment are outlined in Table 2. Two sets of single-pass multilayer additive manufacturing tests were conducted, designated as UIT-WAAM (with ultrasonic impact) and NO-UIT-WAAM (without ultrasonic impact).

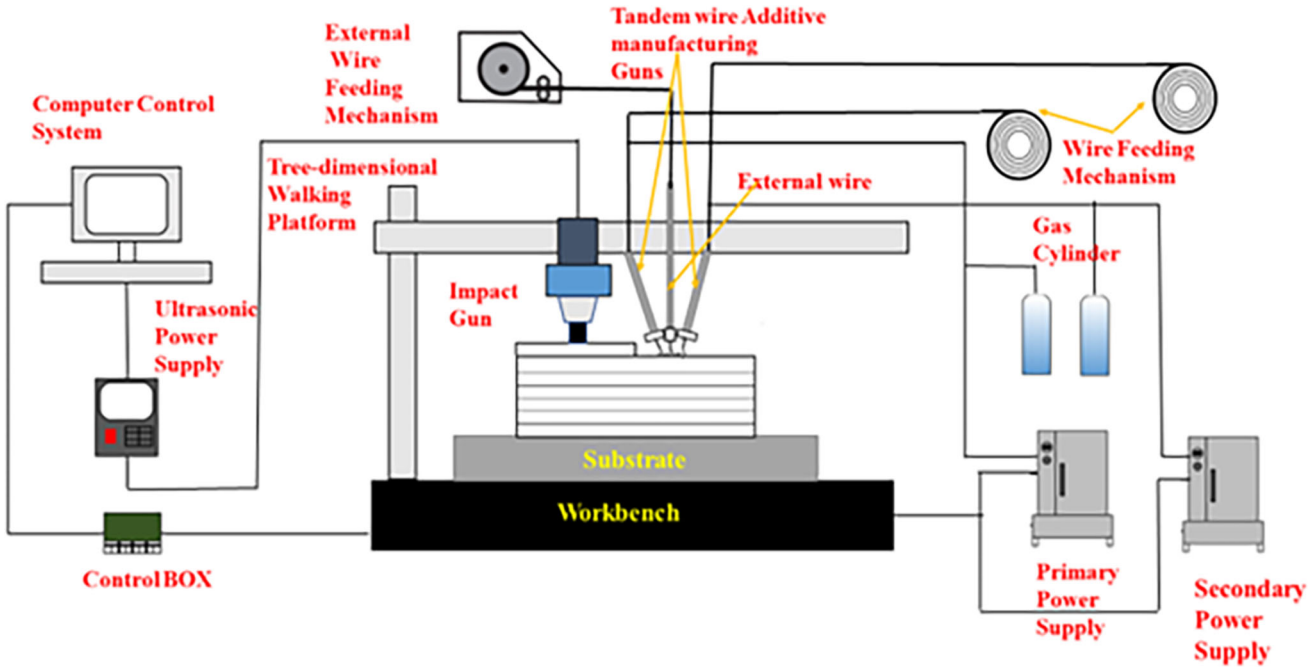


Fig. 1 Schematic diagram of experimental system

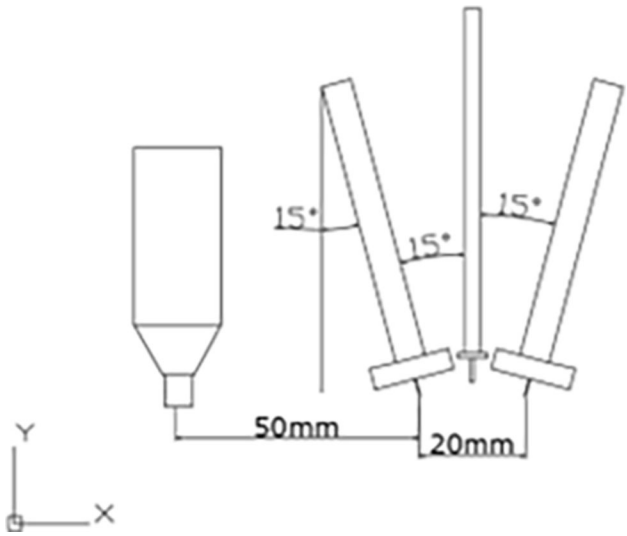


Fig. 2 Additive manufacturing guns and ultrasonic impact gun arrangement diagram

For the single-layer multi-pass samples, metallographic samples are taken from the top, middle, and bottom regions of both the directly deposited parts and the ultrasonic impact parts. These samples are then polished and the surfaces are etched with a 4% nitric acid alcohol solution for 60 s to reveal their microstructure. The phase distribution and microstructure are analyzed using electron backscatter diffraction (EBSD) equipment, and the results are processed with AZtecCrystal software. Vickers hardness measurements are conducted on the samples using a HVS-1000Z automatic turret digital microhardness tester, with a load set to 0.5 kg and a loading time of 10 s. Hardness variations in the middle region are measured every 1 mm along both sides of the centerline of the vertical cross section, starting from the bottom. Tensile tests are conducted using an SHT4106 electronic universal testing machine to assess the tensile mechanical properties of two groups of single-layer multi-pass formed parts. The sampling positions for the tensile specimens and their dimensions are shown in Fig. 3. The tensile specimens are taken in two directions: along the deposition direction (vertical) and perpendicular to the deposition direction (horizontal), with four samples taken in

Table 1 Chemical composition of wire and substrate materials (Mass Fraction, wt.%)

Material	C	Mn	P	S	Si	Cr	Ni	Mo	N	Fe
Wire	0.012	2.18	0.019	0.01	0.57	18.97	12.7	2.31	0.039	Balance
Substrate	0.057	0.74	0.021	0.002	0.36	18.18	8.04	…	…	Balance

Table 2 Process parameters of WAAM

	Tandem wire feeding speed, m/min	External wire feeding speed, m/min	Travel speed, mm/s	Gas flow rate, L/min	Ultrasonic vibration frequency, KHz	Ultrasonic power, W	Interlayer temperature, °C
No-UIT-WAAM	6	6	5	15	0	0	150
UIT-WAAM	6	6	5	15	40	2000	150

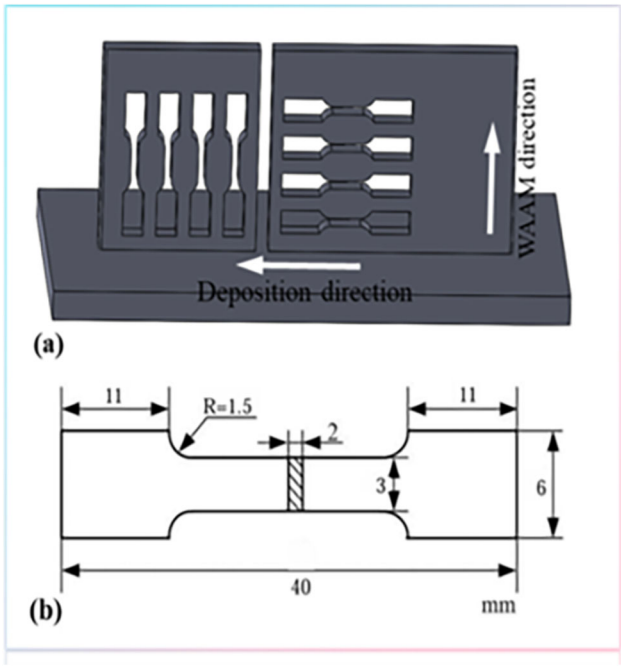


Fig. 3 Single-layer multi-pass test of 316L stainless steel. (a) Sampling positions (b) Dimensions of tensile specimens

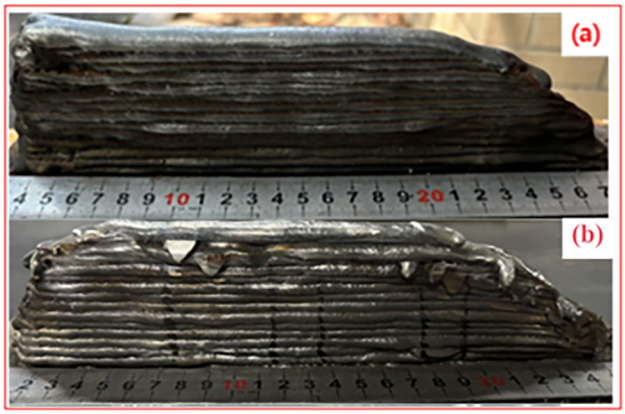


Fig. 4 Macroscopic morphology image of single-layer multi-samples. (a) NO-UIT-WAAM (b) UIT-WAAM

each direction, totaling eight samples. Finally, the deposition efficiency is calculated and compared with other additive manufacturing methods for 316L stainless steel.

3. Results and Discussion

3.1 Macroscopic Morphology of the Formed Components

Figure 4 exhibits samples produced through single-pass multilayer arc additive manufacturing, with both samples featuring generally smooth side surfaces. However, as depicted in Fig. 4(b), the UIT-WAAM sample demonstrates molten pool flow at higher layer counts. This occurs due to the worsening of cooling conditions with an increasing number of layers, which leads to an expansion of the liquid molten pool and a prolongation of its dwell time, making it more prone to flow

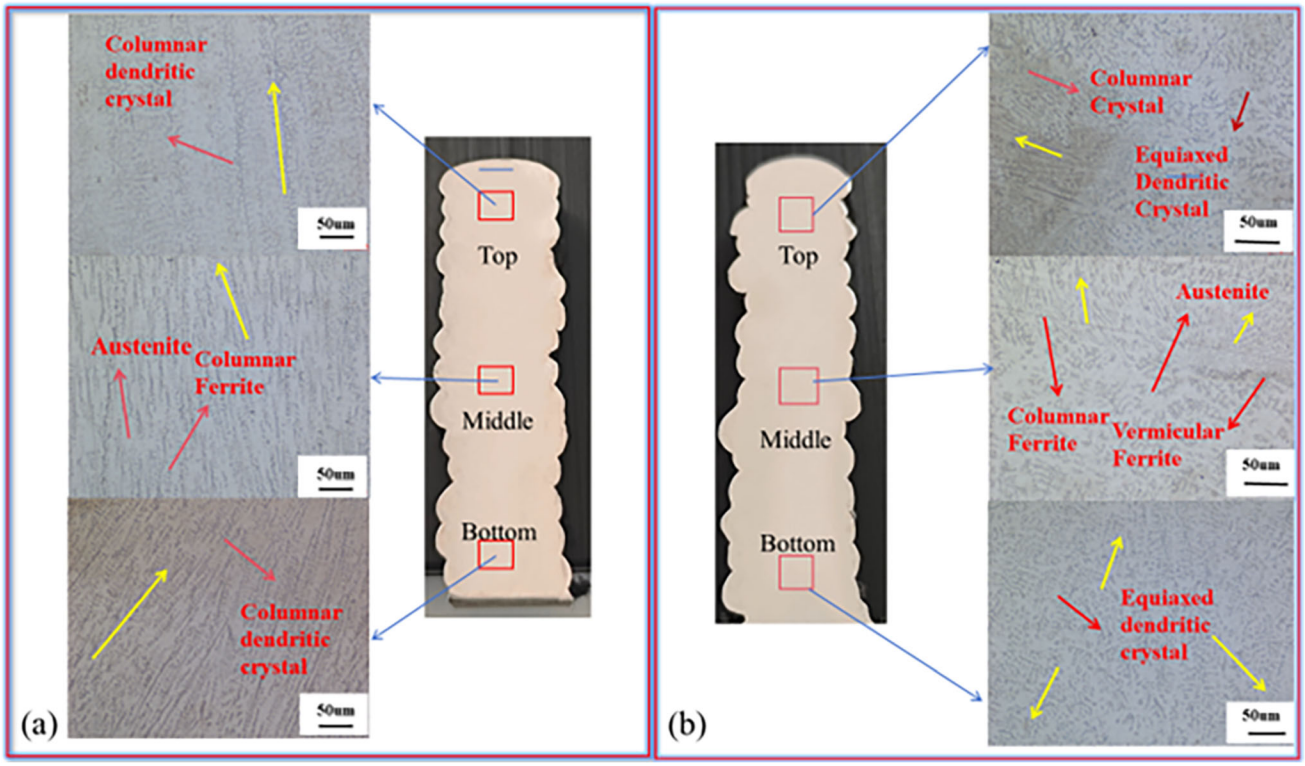


Fig. 5 Metallographic micrographs of single-channel multilayer samples. (a) NO-UIT-WAAM (b) UIT-WAAM

under the influence of ultrasonic impact. Additionally, both samples display notable metal flow at the arc extinguishing end as a result of co-directional stacking.

3.2 Microstructure of the Formed Components

Figure 5 displays the microstructures at different locations of the NO-UIT-WAAM and UIT-WAAM samples, respectively. The microstructure of the deposited samples consists primarily of austenite with a small amount of ferrite. The microstructure of the NO-UIT-WAAM specimen mainly features elongated columnar dendritic crystal, with the shape of the ferrite predominantly being columnar; in contrast, the UIT-WAAM specimen exhibits finer columnar dendritic crystal and equiaxed dendritic crystal, with ferrite predominantly having columnar and vermicular shapes. Notably, there are many fine columnar dendritic crystals near the fusion line, which is attributed to the significant undercooling of the upper weld bead in that region during the additive manufacturing process, preventing the grains from growing sufficiently in size. As illustrated in Fig. 5(b), the grains in the UIT-WAAM specimen fragment under the impact effect, resulting in a refined microstructure with uniform grain sizes near the fusion line. This refinement can be attributed to cavitation effects (Ref 35) and acoustic flow (Ref 36) within the molten pool, both induced by ultrasonic impact, which stimulate nucleation and grain fragmentation during solidification. These mechanisms significantly alter grain morphology. Furthermore, ultrasound intensifies fluid flow within the molten pool (Ref 37), distributing cavitation-generated nuclei and fragmented dendritic crystals evenly, thereby enhancing the formation and survival of equiaxed nuclei. The increased flow rate also improves heat exchange, leading to a more uniform temperature distribution that reduces temperature gradients and columnar crystal growth. These combined mechanisms promote the transformation of columnar crystals into equiaxed crystals within the molten pool.

Additionally, Fig. 5(a) shows that the microstructure of the NO-UIT-WAAM specimen varies, with finer columnar dendritic crystals at the bottom due to direct contact with the substrate and higher cooling rates. As the height of the single-channel multilayer part increases, the cooling rate decreases due to reduced heat exchange, shifting the primary heat transfer mode from conduction (to the substrate) to radiation and convection (to the air) (Ref 38). This results in heat accumulation within the part. The maximum cooling rate is perpendicular to the substrate, aligning with the grain growth direction. Without external forces, grains in the middle and top regions develop into coarse columnar grains. Grain growth in the NO-UIT-WAAM specimen follows the deposition direction; however, ultrasound application suppresses grain growth in the WAAM direction and deflects it toward the ultrasonic impact direction. This deflection aligns with the heat flow direction, inhibiting grain growth along the WAAM direction during solidification and leading to growth in various directions. The yellow arrows in the figure indicate the crystal growth direction.

As shown in Fig. 6(a1) and (a2), the phase diagram of the two samples indicates that the area is almost entirely austenite, with the ferrite content approaching 0. This is because the EBSD scanning area is relatively small and the ferrite content in the stainless steel is low, resulting in an almost negligible ferrite content in this region. As shown in Fig. 6(b_1) and (ba_2), based on the IPF+GB diagrams of both samples, it can be observed that the grain orientations in the center of the NO-UIT-WAAM specimen are predominantly oriented along the $<101>$ and $<001>$ directions, with almost no $<111>$ orientation. The overall grain orientation of the UIT-WAAM specimen is similar

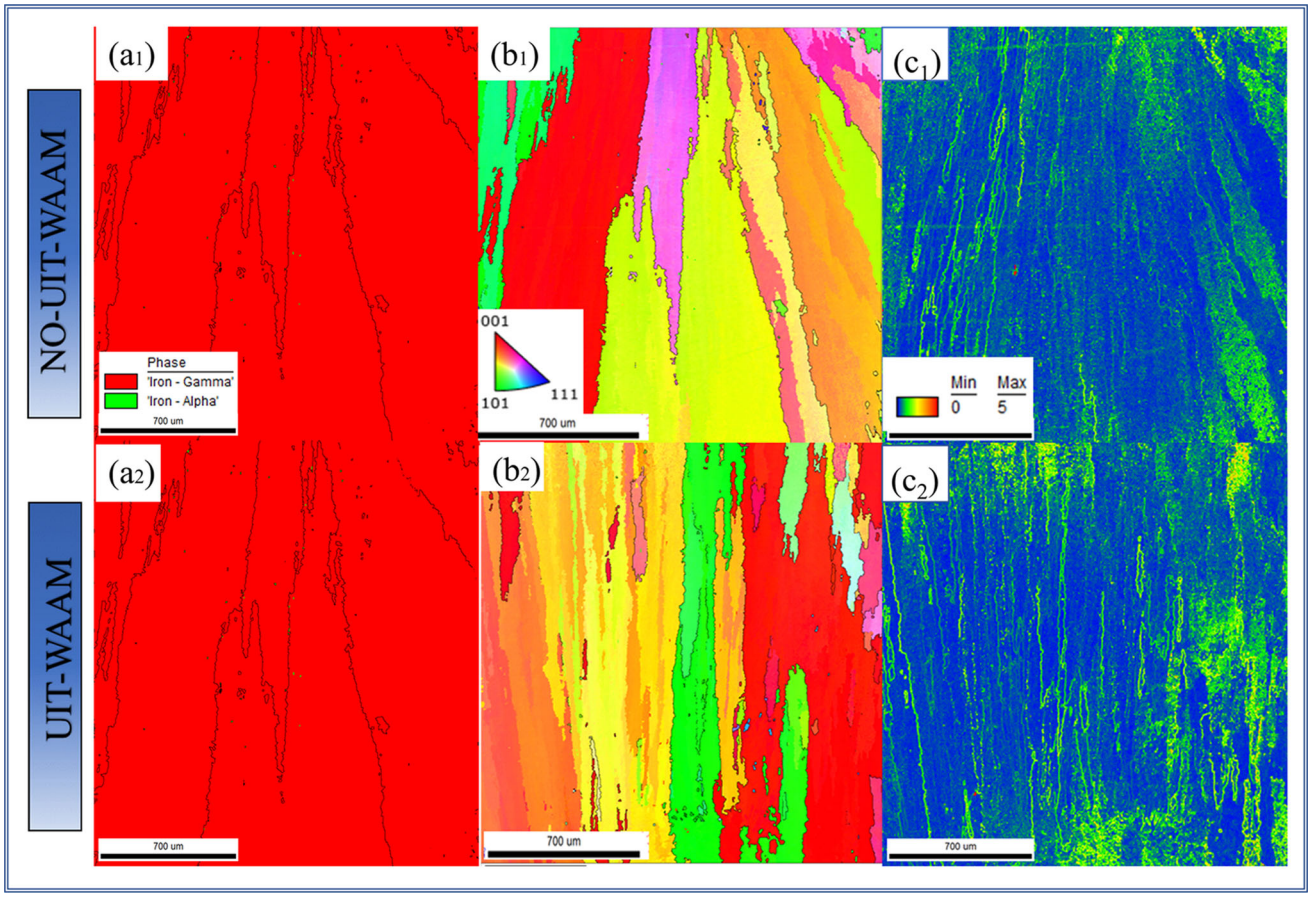


Fig. 6 EBSD test results of NO-UIT-WAAM specimen and UIT-WAAM specimen. (a_1-a_2) Phase maps of NO-UIT-WAAM specimen and UIT-WAAM specimen. (b_1-b_2) IPF+GB maps of NO-UIT-WAAM specimen and UIT-WAAM specimen. (c_1-c_2) KAM maps of NO-UIT-WAAM specimen and UIT-WAAM specimen

to that of the directly deposited samples, indicating that ultrasonic impact did not change the grain orientation. However, it did lead to a certain refinement in the morphology of the austenitic grains. Using the AztecCrystal software, the number and total length of the grain boundaries were counted. The total number of grain boundaries in the NO-UIT-WAAM specimen was 25,962, with a total length of 8.75 cm. In contrast, the total number of grain boundaries in the UIT-WAAM specimen was 57,175, with a total length of 18.87 cm. Compared to the NO-UIT-WAAM specimen, the UIT-WAAM specimen exhibited a higher number of grain boundaries and a greater total length of grain boundaries. As a result, the mechanical properties of the UIT-WAAM specimen are superior to those of the NO-UIT-WAAM specimen. This improvement is attributed to the greater number of grain boundaries, which provides a stronger barrier to dislocation movement, making it more difficult for the material to undergo plastic deformation (Ref 39), thereby enhancing the material perform.

The local misorientation distribution at grain boundaries, known as kernel average misorientation (KAM), is commonly used to analyze the grain orientation and mismatch of crystal materials. KAM_{ave} (average K value) is closely related to geometrically necessary dislocations (GNDs) and represents the average orientation error between local grains. From the KAM map, the geometric dislocation density can be quantitatively calculated, reflecting the uniformity of plastic deformation within the material. A larger KAM value in the sample indicates a greater degree of plastic deformation or defect density at that location (Ref 40). As shown in Fig. 6(b_1) and (b_2), in the KAM image, a higher density of green areas indicates a larger KAM value, the average K value of the NO-UIT-WAAM specimen is approximately 0.498°, while the average K value of the UIT-WAAM specimen is approximately 0.571°. GNDs are formed due to inhomogeneous deformation and can be calculated using the following formula:

$$\rho_{GND} = \frac{2KAMave}{ub} \quad \text{(Eq 1)}$$

Here, μ is the step size of the EBSD experiment (3.3 μm), b is the Burgers vector (2.35×10^{-10}), which ρ_{GND} represents the density of intergranular slip dislocations. The calculated values for the two parameters are $1.28 \times 10^{15}/m^2$ and $1.47 \times 10^{15}/m^2$, respectively. lSo the ρ_{GND} of the UIT-WAAM specimen is significantly higher than that of the NO-UIT-WAAM specimen, indicating a greater degree of plastic deformation. Larger plastic deformation zones can lead to lattice distortion and the formation of substructures, such as dislocations and low-angle grain boundaries, which in turn increase the number of dislocations. The increase in dislocation density will hinder the movement and slip of dislocations. Additionally, the increased dislocations can serve as 'fixed points' for grain boundaries, facilitating their movement and the formation of new grain boundaries. Meanwhile, a higher KAM value indicates that more energy is stored in the corresponding

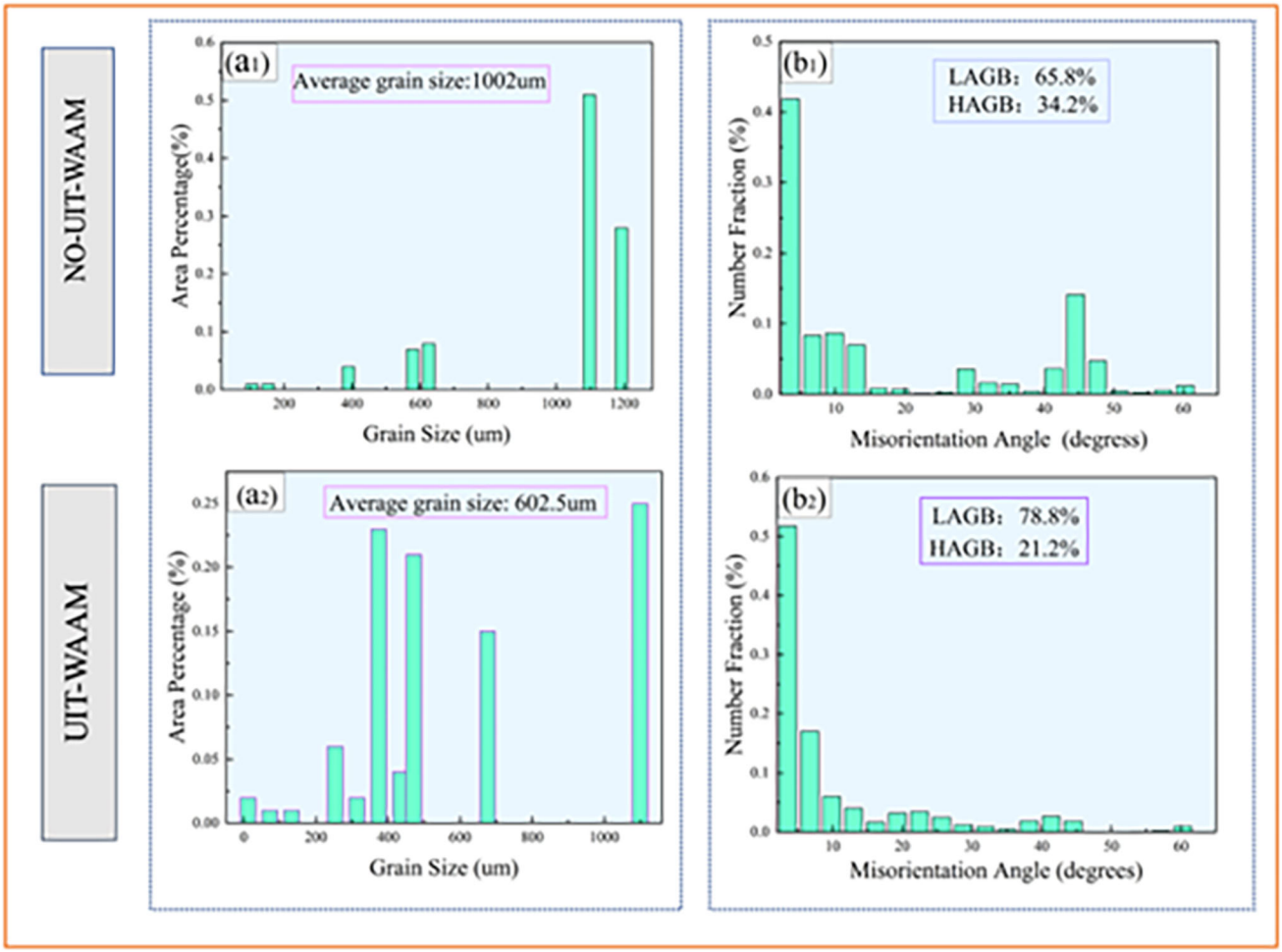


Fig. 7 Grain size statistics charts and orientation deviation angle maps of NO-UIT-WAAM specimen and UIT-WAAM specimen. (a_1-a_2) Grain size statistics charts of NO-UIT-WAAM specimen and UIT-WAAM specimen. (b_1-b_2) The orientation deviation angle maps of NO-UIT-WAAM specimen and UIT-WAAM specimen

Table 3 Comparison of heat input of different approaches to arc additive manufacturing of 316L stainless steel

Reference	Method	Heat input, J/mm
This paper	UIT+ tandem wire with external wire	2273.6
(Ref 43)	CMT+P	263.2
(Ref 44)	PAP-WAAM	566.8
(Ref 45)	CMT	150.5

position (Ref 41), increased the driving force for recrystallization. Therefore, areas with a high dislocation density are more likely to undergo recrystallization, which is also the reason why fine grains are generated at the grain boundaries of ultrasonically impacted materials, thereby increasing the hardness and strength of the material to some extent.

It is well known that the grain size of materials has a significant impact on their plastic strength. Materials with smaller grain sizes typically exhibit higher strength and hardness. This is because finer grains impede dislocation movement and slip at the grain boundaries, increasing the material's resistance to deformation and thus enhancing its properties (Ref 42). In the observations shown in Fig. 7(a_1) and (a_2), the average grain size of the UIT-WAAM specimen is 602.5μm, while the grain size of the NO-UIT-WAAM specimen is 1002μm. Both samples display relatively coarse grains, which is due to the use of external wire feeding-assisted tandem wire–arc additive manufactured 316L stainless steel in this study, resulting in a large heat input.

The formula for calculating welding heat input is given by the relationship:

$$Q = \frac{U * I}{v} \quad \text{(Eq 2)}$$

where the welding voltage U=29.4 V, the welding current I=232 A, and the welding speed v=6 mm/s. The heat input of

one welding torch is 1136.8 J/mm. Due to the presence of two welding guns, therefore, the welding heat input can be obtained as Q=2273.6 J/mm. Table 3 shows the comparison of 316L stainless steel manufactured by arc additive manufacturing under different processes. According to Table 3, the heat input of this method is higher than that of other methods. This leads to slow cooling during the additive process and excessive heat, causing the grains to continuously grow and resulting in larger grain sizes. However, it is noteworthy that the average grain size of the UIT-WAAM specimen is finer compared to that of the NO-UIT-WAAM specimen. Moreover, grains larger than 600μm account for 87% of the NO-UIT-WAAM specimen, while only 40.7% of the UIT-WAAM specimen fall into this category. This indicates that the ultrasonic impact samples have a certain grain refinement effect compared to the directly deposited ones. This is due to the grain refinement effect of UIT, which can lead to grain boundary strengthening. The mechanism behind this is that the irregular arrangement of atoms at the grain boundaries hinders dislocation movement. The strengthening effect can be expressed by the Hall–Petch relationship (Ref 5):

$$\delta_y = \delta_{ss} + kd^{\frac{1}{2}} \quad \text{(Eq 3)}$$

where δ_y is the yield stress and d is the grain size. It can be observed that the smaller the grain size, the more pronounced the grain boundary strengthening effect.

However, because the ultrasonic impact is applied to the surface and its effective depth is relatively shallow, some energy is dissipated when the vibration propagates from the high acoustic impedance solid to the low acoustic impedance liquid (Ref 46). Therefore, the effectiveness of ultrasonic impact in refining grains needs to be improved.

Figure 7(b_1) and (b_2) shows the grain orientation angle distribution of the directly deposited parts and the ultrasonic impact parts. The orientation difference of 2° to 15° is classified as low-angle grain boundaries (LAGB), while that greater than 15° is classified as high-angle grain boundaries (HAGB). Among them, the low-angle grain boundaries in the NO-UIT-WAAM specimen account for 65.8%, and the high-angle grain boundaries account for 34.2%. In contrast, the low-angle grain boundaries in the UIT-WAAM specimen account for 78.8%, while the high-angle grain boundaries account for 21.2%. The proportion of low-angle grain boundaries in the UIT-WAAM specimen is higher. This is because the Read–Shockley model indicates that low-angle grain boundaries are composed of dislocations (Ref 47). The effect of ultrasonic impact leads to plastic deformation of the deposited layer, which in turn increases the dislocation density in the microstructure of the deposited layer. Under the interlayer thermal effects of additive manufacturing, some substructures (dislocations and sub-grains) remain within the deposited layer. The interactions between dislocations hinder dislocation slip and plastic deformation, resulting in increased strength and hardness of the material. At the same time, the smaller orientation differences of low-angle grain boundaries and their greater quantity contribute to a more uniform crystal structure, thereby reducing stress concentration and further enhancing the performance of the UIT-WAAM specimen.

The pole figure of its austenitic matrix is shown in Fig. 8. It can be observed that the maximum density of texture orientation for the NO-UIT-WAAM specimen is 35.18, while the maximum density for the UIT-WAAM specimen is 14.37. The pole figure of the NO-UIT-WAAM specimen is almost symmetrical around the X0 and Y0 axes, indicating a distinct

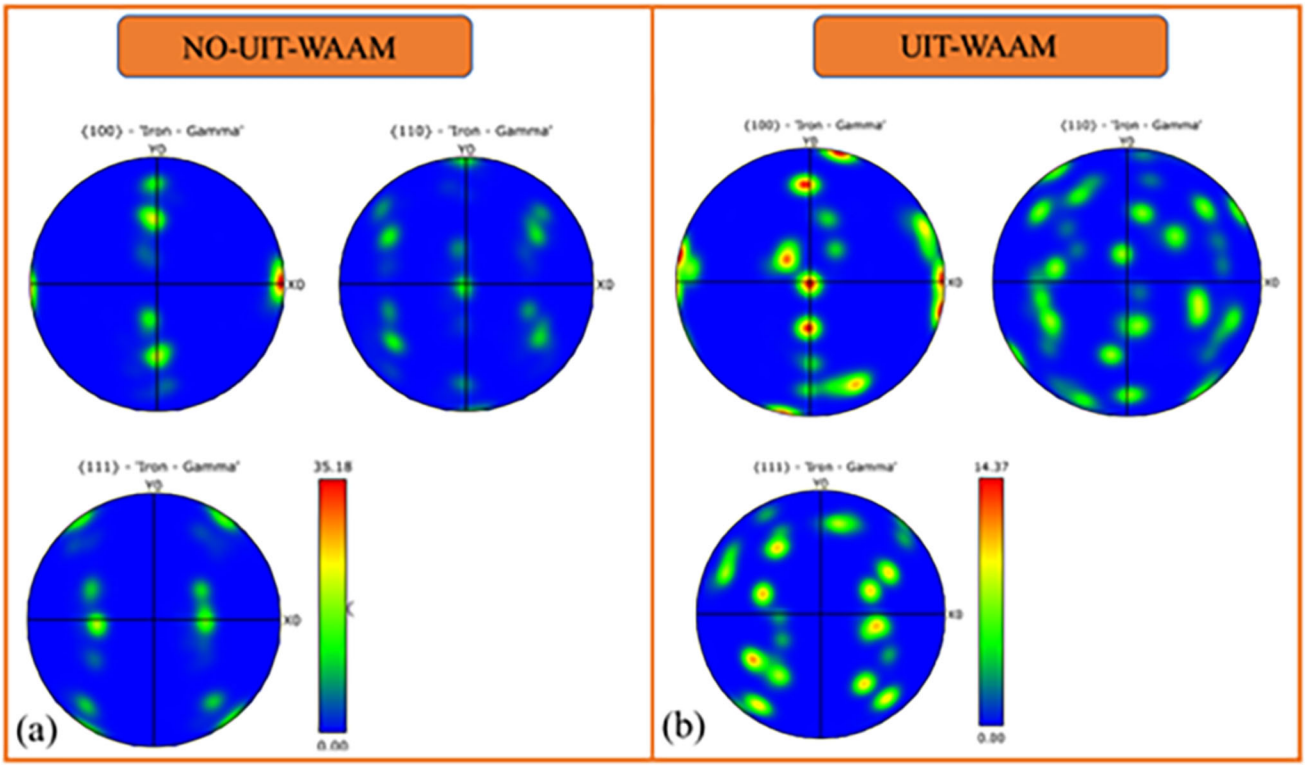


Fig. 8 Pole figures of single-channel multilayer samples. (a) NO-UIT-WAAM (b) UIT-WAAM

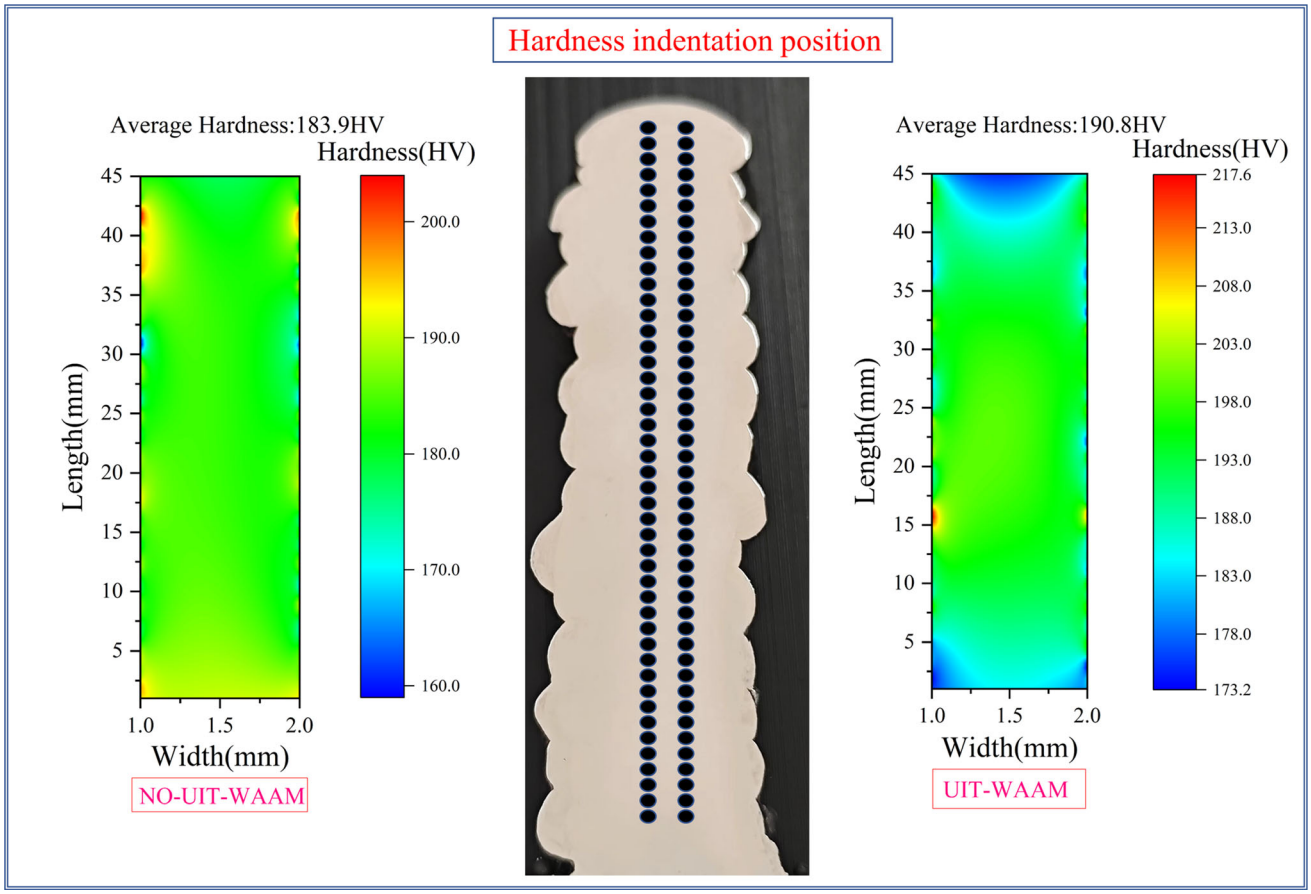


Fig. 9 Microhardness of ultrasonic impact and external wire feeding-assisted tandem wire–arc additive manufactured 316L stainless steel.

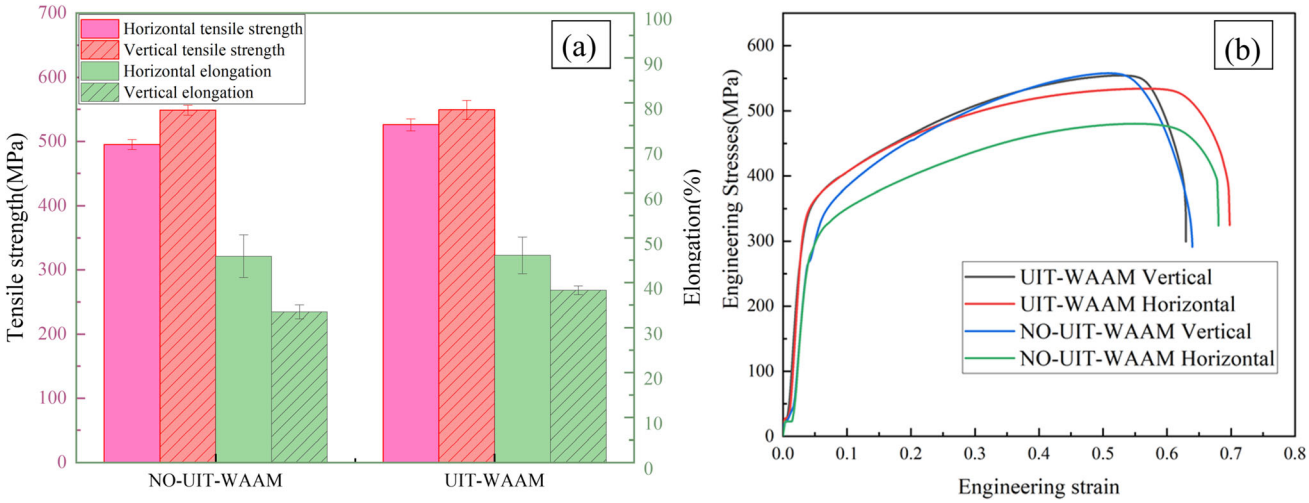


Fig. 10 Mechanical properties diagram of ultrasonic impact and external wire feeding-assisted tandem wire–arc additive manufactured 316L stainless steel. (a) Tensile strength and elongation. (b) Stress–strain curve

texture direction within. In contrast, the orientation of the UIT-WAAM specimen shows a random distribution in the pole figure, with less distinct texture direction. It is well known that a stronger texture orientation indicates severe anisotropy and uneven deformation during the tensile testing process. Therefore, from the above analysis, it can be concluded that there is a significant texture in the top layer of the NO-UIT-WAAM specimen, while the application of ultrasonic impact can suppress the growth of columnar grains in its microstructure, thereby reducing the preferential orientation of the sample and decreasing anisotropy.

3.3 Mechanical Properties of the Formed Component

3.3.1 Hardness. After measurement and plotting, the microhardness distribution maps of the NO-UIT-WAAM specimen and the UIT-WAAM specimen at each point were obtained, as shown in Fig. 9. The hardness of the NO-UIT-WAAM specimen concentrated between 175HV and 190HV, while the UIT-WAAM specimen impact parts concentrated between 180HV and 200HV. The maximum and minimum Vickers hardness values for the NO-UIT-WAAM specimen were 204HV and 159HV, respectively, with an average value of 183.9HV; for the UIT-WAAM specimen, the maximum and minimum Vickers hardness values were 217.8HV and 173.2HV, respectively, with an average value of 190.8HV. This indicates that the hardness of single-pass multilayer formed parts with ultrasonic impact assistance has a certain improvement in both average, maximum, and minimum hardness compared to those without ultrasonic impact, and the range of fluctuation is smaller. This provides evidence that ultrasonic impact technology can enhance the hardness of single-pass multilayer formed parts, having a positive impact on arc additive manufacturing production.

3.3.2 Tensile Properties. As shown in Fig. 10, the horizontal tensile strength of the NO-UIT-WAAM specimen is 495 ± 7.8 MPa, and the vertical tensile strength is 548.7 ± 7.7 MPa. The horizontal tensile strength of the UIT-WAAM specimen is 526 ± 9.4 MPa, while the vertical tensile strength is 549.3 ± 14.7 MPa. The horizontal elongation of the NO-UIT-WAAM specimen is 45.9% ± 4.1%, and the vertical elongation is 33.55% ± 1.5%.The horizontal elongation of the UIT-WAAM specimen is 46.1 ± 4.7%, and the vertical elongation is 38.3 ± 0.9%. Both the NO-UIT-WAAM and the UIT-WAAM specimen show that the horizontal tensile strength is slightly lower than the vertical tensile strength; however, the horizontal elongation is slightly higher than the vertical elongation, indicating significant anisotropy in tensile properties in different orientations. The differences in performance in the two directions can be attributed to two reasons. On the one hand, during the additive manufacturing process, there is directional grain growth along the deposition direction, leading to uneven mechanical properties in the samples. On the other hand, microscopic analysis reveals the presence of pore defects in the interlayer regions of the samples. According to the research by Ma et al. (Ref 48), internal pores reduce the load-bearing area, promoting the propagation of microcracks during tensile testing and even leading to localized brittle fractures, which affects the mechanical properties.

Compared to direct deposition, the application of ultrasonic impact assistance enhances both the horizontal and vertical tensile strengths, while the difference between the two tensile strengths decreases, indicating reduced anisotropy. Meanwhile, the elongation increases slightly compared to the directly deposited samples. This improvement is attributed to the fact that the internal microstructure of the formed part transitions from coarse columnar grains to finer columnar grains with the introduction of ultrasonic impact assistance. According to the Hall–Petch theory, smaller grain sizes lead to higher strength in the formed parts, so incorporating ultrasonic impact assistance results in an improvement in the tensile mechanical properties of the formed parts. However, the enhancement of mechanical properties in the printed parts due to ultrasound is minimal, with hardness, tensile strength, and elongation exhibiting only slight improvements. The reason for the small improvement in mechanical properties is that the thermal input from the tandem wire and arc additive manufacturing is relatively large, resulting in predominantly coarse columnar grains. Although ultrasonic impact has a certain grain refinement effect, its impact is not particularly significant. Therefore, the enhancement of mechanical properties due to ultrasonic impact is not sufficiently pronounced in this study. In future research, we will alter the vibration frequency of the ultrasonic impact or modify the application method to further enhance the influence of ultrasonic impact on mechanical properties, thereby improving the microstructural and properties of the formed parts.

Figure 11(a) and (b) shows the typical fracture morphologies in the horizontal and vertical directions for NO-UIT-WAAM specimen. From the microscopic appearance of the fracture surfaces, it can be distinguished that both the formed parts without ultrasonic impact and those with ultrasonic impact exhibit numerous dimples on their tensile fracture surfaces, indicating typical ductile fracture characteristics. Figure 11(c) and (d) illustrates the typical fracture morphologies in the horizontal and vertical directions for the UIT-WAAM specimen. A comparison reveals that the dimples in the ultrasonic impact samples are larger, deeper, and more uniformly distributed, with most sizes concentrated below a few microns. In contrast, the dimples in the directly deposited samples are smaller, shallower, and more dispersed. There is a positive correlation between the size of the dimples and the plasticity of the material; the larger and denser the dimples, the greater the material's ductility (Ref 49). Therefore, compared to the NO-UIT-WAAM specimen, the UIT-WAAM specimen exhibit better elongation, likely due to the smaller grain size of the samples subjected to ultrasonic impact, which results in smaller pore sizes generated during the tensile process. Additionally, the vertical fracture dimples in the formed parts are relatively small, shallow, and sparsely distributed, while the horizontal fracture dimples are larger, deeper, and more densely distributed. This indicates that the horizontal formed parts have better elongation, which strongly supports the conclusions drawn from the aforementioned tensile tests.

3.4 Deposition Efficiency

Deposition efficiency is defined as the wire feed rate per hour. Let deposition efficiency be represented by the letter A, with the following calculation formula:

$$A = (v_1 + v_2 + v_3) * n * r^2 * \rho \quad \text{(Eq 4)}$$

where v_1 and v_2 represent the feed rates of the dual wires, v_3 represents the feed rate of the externally fed wire, r is the wire radius, and ρ is the wire density. This study uses 316L stainless steel as the test material, with a wire radius (r) of 0.6 mm and a density of 7.98 g/cm^3. The feed rate for both wires is set at 6 m/min, with the externally fed wire also at 6 m/min. After calculations, we find that the efficiency (A) for the tandem wire system with externally fed wire for 316L stainless steel is approximately 9.75 kg/h, meaning that this method can deliver 9.75 kg of welding wire per hour. Table 4 presents a comparison of the efficiency and performance of different approaches to arc additive manufacturing of 316L stainless steel.

From Table 3, it can be seen that the efficiency of the single-pass multilayer sample produced by the ultrasonic impact and external wire feeding-assisted tandem wire–arc additive manufacturing (WAAM) method reaches 9.75 kg/h, which is

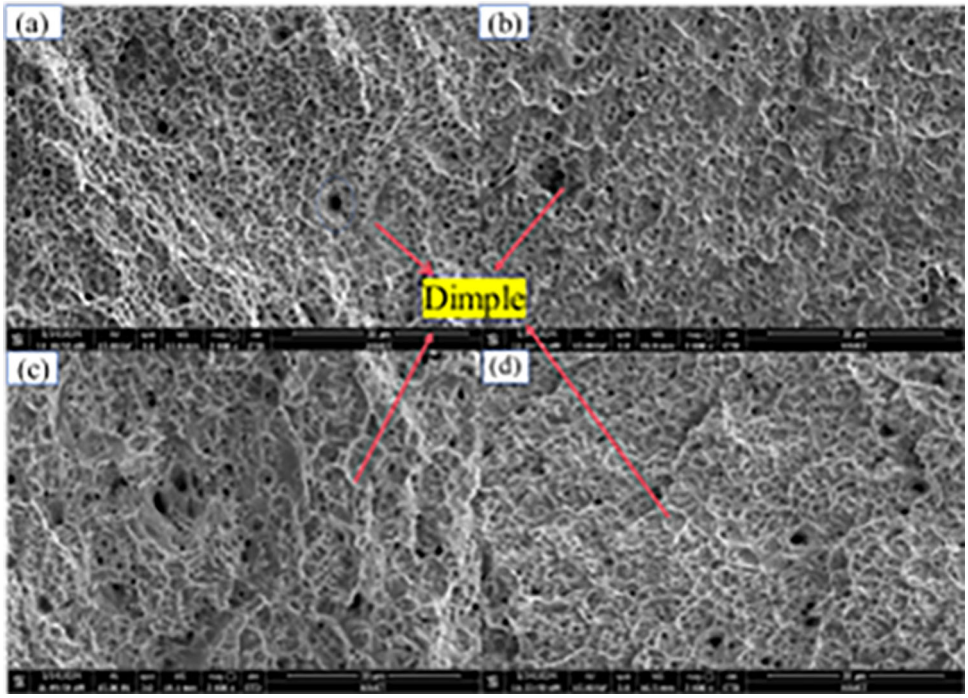


Fig. 11 Microscopic morphology of fracture surfaces in ultrasonic impact and external wire feeding-assisted tandem wire–arc additive manufactured 316L stainless steel. (a) NO-UIT-WAAM horizontal fracture surface. (b) NO-UIT-WAAM vertical fracture surface. (c) UIT-WAAM horizontal fracture surface. (d) UIT-WAAM vertical fracture surface

Table 4 Comparison of efficiency and performance of different approaches to arc additive manufacturing of 316L stainless steel

Reference	Method	Deposition efficiency, kg/h	Hardness (HV)	Tensile strength, MPa		Elongation, %	
				Horizontal	Vertical	Horizontal	Vertical
This paper	UIT+ tandem wire with external wire	9.75	190.8	526	549.3	46.1	38.3
(Ref 50)	CMT+P (single wire)	4.01	174.6	496.7	531.5	…	…
(Ref 51)	GMAW (single wire)	3.25	173	567	595	40	39
(Ref 52)	GMA-AM (single wire)	5.42	200	533		48	

significantly higher than that of the other methods. Meanwhile, in terms of mechanical properties, the average hardness of the single-pass multilayer samples produced by the ultrasonic impact and external wire feeding-assisted tandem wire–arc additive manufacturing reaches 190.8 HV, showing an increase compared to the aforementioned methods, with tensile strength and elongation rates being nearly comparable. This further demonstrates the superiority of the tandem wire–arc additive manufacturing method assisted by ultrasonic impact and external wire feeding.

4. Conclusion

This study investigated the effects of ultrasonic impact and external wire feeding on the microstructure and mechanical properties of 316L stainless steel in arc additive manufacturing. We reached the following conclusions:

1) The ultrasonic impact plus external wire feed assisted tandem wire–arc additive manufacturing platform has been built and a single multilayer specimen has been successfully added, and the deposition efficiency under this process reaches 9.75 kg/h.
2) After applying ultrasonic impact assistance, the microstructure of the samples changed from coarse columnar dendrites to finer columnar dendrites and equiaxed dendrites. Additionally, ultrasonic impact increases the number of grain boundaries and dislocation density in the samples, which impedes dislocation movement and sliding, thereby enhancing the mechanical properties of the samples.
3) The average hardness of the UIT-WAAM specimen is 190.8 HV, while the average hardness of the NO-UIT-WAAM specimen is 183.9 HV, indicating a feeble enhancement in the hardness of the formed parts. After applying ultrasound, the transverse average tensile strength of the single-pass multilayer specimens in-

creased from 495 ± 7.8 MPa to 526 ± 9.4 MPa, and the longitudinal average elongation rate improved from 33.55% ± 1.5% to 38.3% ± 0.9%.

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